

Designing for Trust and Well-being: Identifying Design Features of Highly Automated Vehicles

Vanessa Sauer

Institute of Industrial Engineering and Ergonomics,
RWTH Aachen University
Aachen, Germany
v.sauer@iaw.rwth-aachen.de

Jens Heitland

AUDI AG
Ingolstadt, Germany

Alexander Mertens

Institute of Industrial Engineering and Ergonomics,
RWTH Aachen University
Aachen, Germany

Verena Nitsch

Institute of Industrial Engineering and Ergonomics,
RWTH Aachen University
Aachen, Germany



Figure 1: Exemplary design features of a highly automated vehicle that we considered in our study. Copyright by schneller-vorlauf gmbh.

ABSTRACT

Vehicle automation is one of the major trends in the automotive industry and beyond. In our study, we investigate how future users with different levels of initial trust evaluate design features of level 4 automated vehicles in regards to the features' ability to create passenger well-being. For this purpose, we identified potential design features from existing automated vehicle concepts and asked experts ($n = 15$) to rate them regarding their relevance to passenger well-being. In a second step, we conducted a user study ($n = 69$) to investigate how future users classify those features deemed relevant by the experts. Using the Kano method, the subsample with low

initial trust rated 14 of 28 features as relevant, while the subsample with high initial trust rated 20 of 28 features as relevant. Further, the results indicate that the features deemed important for passenger well-being differ depending on the level of initial trust.

CCS CONCEPTS

• Human-centered computing → Empirical studies in HCI.

KEYWORDS

Passenger well-being, Trust, Vehicle design, Automated driving

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1 INTRODUCTION

Vehicle automation has become an established aspect of automotive companies' strategy and development efforts, including manufacturers, suppliers, and service providers [31]. These efforts include the development of highly automated vehicles (level 4) [27]. Compared to vehicles with lower levels of automation, level 4 automated vehicles promise a paradigm change by relieving the driver from the responsibility of acting as a fall-back. Instead, drivers can engage in more elaborate non-driving related tasks (NDRTs) without being interrupted by frequent take-over requests. This paradigm change entails changed user behavior of the driver, which opens up the opportunity to rethink the vehicle design and make highly automated vehicles less driver-centric and more passenger-centric.

Many ideas on how to rethink the design of highly automated vehicles exist already. These ideas include, for example, research efforts investigating improved ways of designing user interfaces [7], dashboards [9], or seat positions [30]. Research also investigates relevant user needs for a positive user experience with automated vehicles [18, 22].

In industry, multiple manufacturers, suppliers, and new players have presented their vision of level 4 automated cars. Contrary to most research endeavors, industry concepts mainly include a holistic vehicle concept instead of focusing on single vehicle features. Yet, most concepts do not emphasize the passenger experience [33]. To fill the gap of a research perspective on holistic level 4 research concepts, we investigate how future users evaluate design features of presented level 4 vehicle concepts regarding the features' contribution to passenger well-being. We focus on the evaluation of the design features regarding passenger well-being (PWB). PWB is closely related to the user experience by taking into account the passenger's perceptions and resulting responses [6] to derive the passenger's current affective state triggered by the travel experience [29].

The experienced level of PWB not only depends on the external factors of the travel experience. It is also shaped by individual characteristics such as psychological, physical, and situational factors [25]. Psychological factors often result from preconceptions and, in turn, can determine passenger's reaction and, ultimately, the level of PWB. One psychological factor that has proven to influence the perception and reaction to automated vehicles is initial trust [10].

Based on this line of arguing, we investigate in this study if different levels of initial trust change how future users evaluate the design features of automated vehicles regarding the expected level of PWB. To address this research question, we extend a previous user study which utilized the Kano method [13] to investigate how future users classify design features of level 4 automated vehicles regarding PWB [28]. The results presented and discussed in this paper investigate how initial trust may shape the expected effect of the design features on PWB.

2 RELATED WORK

Previous research investigating the design of automated vehicles is broadly spread across different perspectives, such as acceptance factors of automated vehicles [36], NDRTs passengers may engage in [11, 22] and specific approaches and suggestions on how to design the automated vehicle itself. Given the research question

of what design features contribute most to PWB, the latter area of research is most relevant.

2.1 Evaluation of design features

Focusing specifically on PWB, prior research has investigated design features of the driving system such as driving behavior and trajectories [32], or design features of the interior such as lighting [5]. In more general terms, research focusing on design features to create a positive user experience varies in the specification. Studies, with a more global nature, address, for example, the design of user interfaces [7], service design for shared autonomous vehicles [16], or the design of automated vehicles in general [18, 21]. As results, these studies propose user needs and requirements to be considered in the respective design. For example, user interfaces should address the needs of feeling safe, pleasure of driving, reduction of perceived travel time, and customization in the design [7]. The service design for shared automated vehicles should include features that create a perception of safety (such as travel information, control in emergencies), allow the utilization of free time while traveling (e.g., entertainment, adjustable seats), and facilitate communication with the automated vehicle (such as interface design, route and destination information) [16]. Related suggestions are also found for a positive user experience with automated vehicles in general (safety, empowerment and interaction style) [21], and for user requirements for fully automated vehicles (system functionality, input control, information display, and in-vehicle space) [18].

More focused studies propose detailed designs of single aspects of the automated vehicle, such as a dashboard designed to reflect friendship between the vehicle and the driver [9]. A different study proposes an interior with rotatable seats and multimodal information to facilitate take-over requests by increasing situational awareness [30].

Overall, research on the evaluation of design features tends to focus on a general view on user needs and requirements for automated vehicles or by suggesting a specific design for single features such as lighting, seating, or user interfaces. However, to our knowledge, a holistic view of design features and their evaluation from a user perspective is lacking.

2.2 Initial trust

Trust is "the attitude that an agent will help achieve an individual's goals in a situation characterized by uncertainty and vulnerability" [17]. Given that the early stages of automated vehicles are marked by uncertainty, trust plays an essential role in building human-automation relationships and creating willingness to use [36]. In these early stages, where highly automated vehicles are only available as prototypes, future users hardly have the option to experience automated driving. Therefore, the initial level of trust prior to using an automated vehicle is relevant. Initial trust results from past experiences made in similar circumstances or second-hand information such as word-of-mouth [17].

The initial level of trust shapes the perception of design features of automated vehicles [10]. For example, lower levels of trust create a higher need for travel-related information [2] and a more negative experience regarding safety, comfort, and satisfaction [10]. Further,

the level of trust also strongly determines the level of perceived control and willingness to handover control [10, 26]. In sum, the findings of prior research suggest, automated vehicles need to be designed with different levels of initial trust in mind to ensure willingness to use, and acceptance of these vehicles [10].

3 METHODS AND MATERIALS

We utilized a multi-step procedure to evaluate how future users with different levels of initial trust evaluate design features of highly automated vehicles regarding PWB. First of all, we identified possible design features from published vehicle concepts. In a second step, we reduced the number of identified design features with the help of an expert survey to keep the number of features investigated with the user study (step 3) manageable.

3.1 Identifying relevant design features

We analyzed vehicle concepts presented in the last five years (2015–2019) from 20 different manufacturers across nine countries to identify design features with potential relevance for PWB. We used press releases, news articles, and pictures to build a list of design features. To be included in the list, the feature needed to be possible in level 4 automated vehicles as opposed to level 3 or below (e.g., rotatable seats), or provide additional benefits due to the automation (e.g., additional displays – they would be possible in lower automation, but provide additional benefit by facilitating NDRTs that become possible in level 4 automation). We did not consider technical feasibility in the selection to avoid unnecessary limitations of the feature space due to potentially false predictions of future technologies. In total, we identified 70 vehicle features in six areas: vehicle architecture, materials, driving dynamics, ambient features, user interaction, health, and wellness (see Figure 1 for a sketch of exemplary features).

3.2 Expert survey

We utilized an expert survey to verify the design features' relevance for PWB with external opinions. Fifteen experts from automotive companies knowledgeable in automated vehicles and their product requirements rated the identified design features regarding their relevance for creating PWB on a scale from 1–3 (1: not relevant for PWB, 2: unsure, 3: relevant for PWB). Further, the experts were able to add additional features they felt were missing. We used the experts' assessments to reduce the 70 design features to a more easily manageable number of features to be evaluated in a user study. For this purpose, we chose a threshold value of 2.5 for the average rating. Based on the utilized scale from 1 to 3, an average rating of 2.5 and above indicates that the experts on average lean toward the feature being relevant for PWB. In total, 23 design features exceeded the set threshold and were deemed interesting for further investigation in the user study. Additionally, five design features were added to the features considered in the user study based on additions made by the experts.

3.3 User study

We conducted a user study to evaluate the 28 resulting design features regarding PWB from a user perspective, considering different levels of initial trust. For this purpose, we adapted the Kano method

[13] in combination with a static level 4 driving simulator. The Kano method is one of the user-centric methods to identify relevant features of future products from a user perspective [1, 31]. It classifies product features based on two dimensions (the physical fulfillment and user satisfaction) into six different categories: Attractive features (A), One-dimensional features (O), Must-be features (M), Indifferent features (I), Reverse features (R), and Questionable features (Q). The classification can provide insights into the relevance of the different product features from a user perspective [13]. We adjusted the Kano questionnaire to integrate PWB as a user evaluation criterion (see [28] for details). Previously, the Kano method has been successfully applied in the automotive domain, for example, in investigating car seats [15], dashboards [34], HMI designs [14], or vehicle design principles [35].

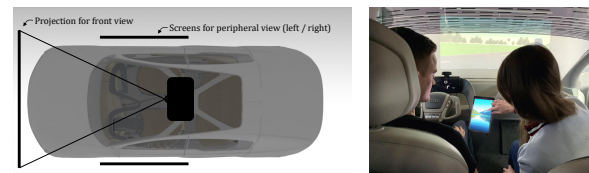


Figure 2: Study setup of static level 4 driving simulator (left) and Kano interview conducted in the simulator (right).

3.3.1 Procedure. The participants were welcomed to the study, gave informed consent, and completed a questionnaire on their demographics. Then they were seated on the driver's seat of the static driving simulator to experience a level 4 automated drive for three minutes (see Figure 2). The drive was without any take-over requests or reliability issues and helped ensure that all participants had the same understanding of level 4 automated driving. After the drive, the participants completed a short questionnaire measuring their level of trust towards automated driving. The participants were instructed to give their general level of trust towards automated driving.

After completing the survey, the experimenter joined the participant in the simulator on the passenger seat and administered the Kano questionnaire as an interview. The definition of passenger well-being and the scale were explained. Participant's understanding was ensured before moving on to the rating of the design features. Each of the 28 design features was explained to the participant utilizing prototypes wherever possible (see Table 1) to ensure a better understanding of the impact the feature could have on PWB. The participant was asked to evaluate the features based on the premises that they used the vehicle privately (no ride-sharing with strangers) and were seated on the driver's seat.

3.3.2 Measures. Initial trust was measured using the German translation [23] of the scale of trust in automated systems [12]. As trust and distrust are hypothesized as distinct dimensions in this scale, only the items measuring trust were collected. Additionally, the participants' interest in cars in general, and their experience with advanced driver assistance systems (ADAS) were measured using a five-point Likert scale (1: very low, 5: very high). The answers to functional and dysfunctional questions for 28 design features using

the adjusted scale from the Kano method were collected (see [28] for details on adjusted scale).

3.3.3 Participants. Participants were recruited from multiple German companies in the automotive sector. Automotive employees were deliberately chosen to ensure that participants have sufficient knowledge of automated driving to be able to estimate the effect of design features on their well-being in future vehicles. The recruitment and study design complied with the specifications and requirements set by the ethics committee of RWTH Aachen University, Germany. In total, 69 participants (23 female, 46 male; aged between 19 and 58, $M = 34.17$ years) took part in the study. Of the sample, 52 participants had a technical background and 17 a non-technical. On average, the sample had a high interest in cars ($M = 4.07$, $SD = .96$) and was experienced with ADAS ($M = 3.83$, $SD = .97$). Regarding trust, the participants' initial trust was, on average, moderately high (scale 1-7, 1: very low, 7: very high, $M = 4.92$, $SD = 1.03$). The participants were organized in two groups based on their initial trust ratings using a median split (high trust: $n = 33$, $M = 5.67$, $SD = .45$, low trust: $n = 36$, $M = 4.14$, $SD = .76$).

3.3.4 Data analysis. We analyzed the collected data with R [24]. We classified the 28 design features from the perspective of the total sample and each trust group using the mode rule if, according to the Fong test, the classification was significant. Else, we used the decision rule to classify the feature as detailed in [3]. Further, we calculated the parameters CS+ and CS-, which provide insights into which features are required to achieve a basic level of PWB ($CS- < -.50$) and which features can create high levels PWB if included ($CS+ > .50$) [20]. To gain further insights into the predictors for trust, we analyzed the demographic data by conducting correlation analysis.

4 RESULTS

The correlation analysis shows that initial trust only correlated significantly with the intent to use automated driving functions ($r = .34$, $p < .001$). There was no significant linear relationship between initial trust and general interest in cars ($r = .07$, $p = .58$) and only a non-significant indication of a linear relationship between initial trust and the level of experience with ADAS ($r = .20$, $p = .09$).

To address the raised research question, the classification of the 28 features of future level 4 automated vehicles is presented in Table 1. Here, the classification and the CS+ and CS- parameters are reported for both trust groups. The results of the total sample are reported as a baseline for comparisons.

5 DISCUSSION

Regarding the predictors of initial trust, only the intent to use automated driving functions correlated significantly with initial trust in our sample. This finding corresponds to the results of other studies, such as [36]. There was no indication for a linear relationship between initial trust in automated vehicles and level of experience with ADAS, or general interest in cars in our sample. However, prior research has found this relationship (e.g., [4, 8]). Reasons may be that we recruited the sample from automotive companies. Therefore, both initial trust and prior experience may be skewed compared to samples from other research.

Viewing the design feature classification, the low trust group views 14 of 28 features (8 Attractive, 3 One-dimensional, 3 Must-be) as relevant for PWB. In comparison, the high trust group classifies 20 of 28 features (12 Attractive, 4 One-dimensional, 4 Must-be) as relevant (see Table 1). When viewing what features were classified as relevant, it appears that for both groups, a strong focus on relaxing is linked to PWB. For example, both groups view a seat that enables a relaxed seating position, has a massage function, and foot rest as attractive. Situational air conditioning, privacy glazing, and themed interiors (e.g., corresponding light, scent, sounds to facilitate relaxing, working, or other NDRTs) are viewed as attractive as well, i.e., these features are not expected but create high levels of PWB. This finding, is also reflected in the CS+ ratings ($CS+ > .50$).

Features that are mandatory to create a minimum level of PWB ($CS- < -.50$, or O and M classifications) include standard features in passenger cars available today (storage, arm rests, vehicle status display, minimum material quality). Further, future users expect a memory function remembering individual settings, and the possibility to adjust the driving style of the automated vehicle to be a minimum requirement. Users from both trust groups have similar expectations on the minimum features required to experience PWB in an automated vehicle, which touch on minimum amenities known from today's cars as well as possibilities to maintain control.

Despite the similar expectations of both trust groups at first glance, the two groups differ in their classification regarding eleven features (highlighted blue in Table 1). For eight features, future users with higher initial trust rate them as attractive or one-dimensional, compared to an indifferent rating from users with low initial trust. The higher relevance of these eight features in the high trust group may be explained in some cases by the fact that users with higher trust may be more willing to give up the monitoring of the vehicle and instead engage in NDRTs. In contrast, lower trust users prefer easy access to the vehicle status [19]. This argument could explain why individual and rotatable seats, active noise cancelling, individual displays, an ambient light displaying the vehicle status, voice control, and a learning system are rated as relevant by high trust users. Low trust users may view these features as indifferent because there is no clear trade-off between feedback and higher comfort.

In contrast, the different categories for the table and the reading light are less plausible. We would expect these features to be more relevant for PWB for high trust users than low trust users if we follow the argument that higher initial trust leads to a more likely engagement in NDRTs. Additionally, there are more design features for which we would have expected differences in the categorization between the two groups. This is especially the case for more safety-critical features that allow control as suggested by [10, 26], such as a flexible steering wheel or the adjustable driving style. We expected that these features would play a more important role for PWB for low trust users. Prior work found that even in vehicles with higher automation levels, future users expect manual control such as an emergency button or a steering wheel [16]. The missing evidence in this study may be related to the fact that participants rated the effect on PWB in general and not during the initial use. Some participants may have made projections into the future and may have assumed that their trust would be higher in the future and blur the distinction between high and low trust groups.

Table 1: Classification of the 28 design features with the Kano method (P: a prototype was used to aid explanations, C: Category, A: Attractive, O: One-dimensional, M: Must-be, I: Indifferent, * indicates significant Fong-test, dark grey bar indicates CS values exceeding thresholds, blue highlight indicates different classifications between trust groups).

Feature	Total Sample			Low Trust			High Trust		
	C	CS-	CS+	C	CS-	CS+	C	CS-	CS+
<i>Driving dynamics</i>									
Adjustable driving style ^P	O	-.55	 .52	O	-.61	 .53	O	-.48	 .52
Active chassis	I	-.13	 .48	I	-.10	 .48	I	-.17	 .48
<i>Vehicle architecture</i>									
Flexible steering wheel ^P	I*	-.32	 .32	I*	-.33	 .30	I*	-.30	 .33
Swivel seats ^P	I*	-.24	 .42	I	-.27	 .42	A	-.21	 .42
Individual seats ^P	O	-.53	 .39	I	-.48	 .29	O	-.58	 .48
Relax position ^P	A	-.38	 .59	A	-.41	 .53	A*	-.34	 .66
Foot rest ^P	A	-.24	 .56	A	-.25	 .56	A	-.22	 .56
Arm rest ^P	M	-.65	 .41	O	-.56	 .36	M	-.76	 .45
Massage ^P	A	-.38	 .68	A	-.42	 .72	A	-.33	 .64
Table ^P	I*	-.22	 .40	A	-.31	 .44	I*	-.13	 .35
Storage ^P	M	-.87	 .41	M	-.83	 .42	M	-.91	 .39
Privacy glass ^P	A	-.14	 .56	A	-.18	 .59	A	-.09	 .53
Cocooning ^P	I*	-.00	 .13	I*	-.00	 .03	I*	-.00	 .23
<i>Materials</i>									
Sustainable materials ^P	I*	-.25	 .45	I*	-.22	 .44	A	-.27	 .45
Material quality ^P	M	-.71	 .41	M	-.69	 .39	M	-.73	 .42
<i>Ambient features</i>									
Active noise cancelling	I	-.12	 .47	I*	-.06	 .39	A	-.18	 .55
Situational air conditioning	A*	-.11	 .64	A*	-.09	 .70	A	-.13	 .58
Situational ambient light ^P	I*	-.07	 .35	I*	-.03	 .33	I*	-.13	 .38
Situational reading light ^P	I*	-.08	 .42	A	-.09	 .56	I*	-.07	 .27
<i>User interaction</i>									
Voice control	I*	-.32	 .36	I*	-.28	 .28	A	-.36	 .45
Own devices ^P	I*	-.17	 .30	I*	-.22	 .42	I*	-.12	 .18
Individual displays ^P	A	-.22	 .47	I*	-.28	 .44	A	-.16	 .50
Vehicle status display ^P	M	-.81	 .42	M	-.83	 .39	M	-.79	 .45
Vehicle status ambient light ^P	I*	-.42	 .42	I*	-.39	 .36	O	-.44	 .47
Memory function ^P	O	-.49	 .57	O	-.50	 .69	O	-.48	 .42
Learning system ^P	A	-.12	 .51	I	-.11	 .47	A	-.12	 .55
<i>Health and wellness</i>									
Relaxation exercise ^P	I*	-.00	 .13	I*	-.00	 .14	I*	-.00	 .12
Themed interiors ^P	A	-.18	 .51	A	-.17	 .47	A	-.19	 .56

6 LIMITATIONS

Our multi-step study approach has some limitations regarding methodology and results. We utilized a static level 4 driving simulator. This may have impaired realism. However, to date, only a few running level 4 automated prototypes are on the roads, making driving simulators the next best option to obtain the most generalizable results. The design features were only demonstrated as prototypes but not experienced over longer periods of time. Thus, this classification is rather to be seen as a first impression and may change if participants can interact with and experience features over longer time periods. Further, we asked participants to evaluate the features assuming they were alone and seated in the driver's

seat. Although this may cover most of today's car driving behavior where many drivers travel on their own, the classification may change when the scenario is changed to ride-sharing or different seats.

Additionally, several features are not yet available today. Consequently, it may be hard for participants to estimate the benefit of these features for PWB, despite having prototypes to illustrate and facilitate evaluation. For that reason, we recruited participants from the automotive sector to ensure a better understanding of automated driving and more accurate predictions. However, this approach to recruiting participants comes at the expense of generalizability of the results to larger populations. We created subsamples

with high and low initial trust through a median split. Thus, trust levels were externally given, as the participants were not specifically recruited depending on their trust levels, nor were trust levels manipulated in the study design. Consequently, the differences between the trust levels were not extreme but sufficient to observe some differences in the evaluation of the design features.

7 CONCLUSION

Based on an exploratory approach, we derived 28 vehicle design features with the potential to create PWB in a level 4 automated vehicle from publicly presented automated vehicle concepts and an expert survey ($n = 15$). In an additional step, we evaluated how 69 future users view these 28 features for creating PWB using the Kano method while considering their initial trust levels. Overall, design features strongly related to relaxation (e.g., a seat offering a relax position, foot rest, and massage function) are considered relevant for PWB. This trend is even stronger for users with higher initial trust ratings. They find design features that enable NDRTs during automated driving highly relevant for PWB, even at the expense of giving up control and feedback channels (e.g., active noise cancelling, swivel seats, voice control, individual displays). On the other hand, users with lower initial trust view fewer design features as relevant for their PWB (14 von 28 features relevant).

These results give an overview of how future users evaluate multiple design features proposed for highly automated vehicles by industry regarding PWB. The results confirm that initial trust plays a role in the perception of vehicle features and, consequently, how PWB during automated driving can be created. These findings can guide the user-centric development of future automated vehicles by taking a holistic view of the design features of said vehicles.

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